

72 A diploid plant cell divides by mitosis.

After mitosis of this cell, a mutation occurs that changes the genotype of **one** of the daughter cells. This mutant daughter cell produces a non-functional enzyme instead of the functional enzyme produced by the other daughter cell. This mutation has no effect on the phenotype of the plant or the number or length of chromosomes in the plant cell.

Which of the following statements describe the daughter cells after the mutation has occurred?

- 1** The chromosomes in the nucleus of each daughter cell will contain the same genes.
 - 2** Both daughter cells will contain the same alleles for every gene.
 - 3** The sequence of bases along each allele will be the same in each daughter cell.
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- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3